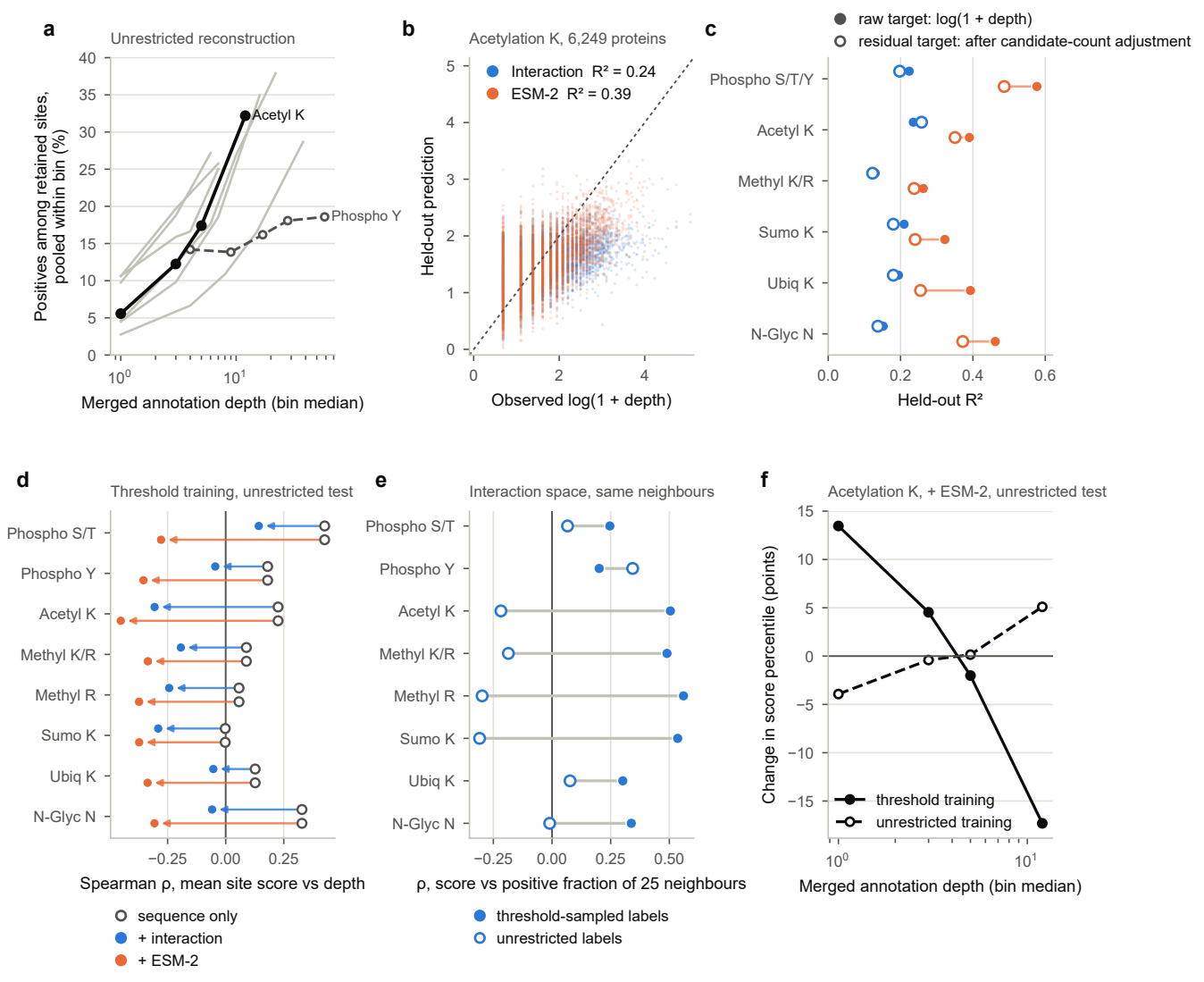

a
Unrestricted reconstruction
Positives among retained sites, pooled within bin (%)
Acetyl K
Phospho Y
Merged annotation depth (bin median)
b
Acetylation K, 6,249 proteins
Interaction R² = 0.24
ESM-2 R² = 0.39
Held-out prediction
Observed log(1 + depth)
c
raw target: log(1 + depth)
residual target: after candidate-count adjustment
Phospho S/T/Y
Acetyl K
Methyl K/R
Sumo K
Ubiq K
N-Glyc N
Held-out R²
d
Threshold training, unrestricted test
Phospho S/T
Phospho Y
Acetyl K
Methyl K/R
Methyl R
Sumo K
Ubiq K
N-Glyc N
Spearman ρ, mean site score vs depth
sequence only
+ interaction
+ ESM-2
e
Interaction space, same neighbours
ρ, score vs positive fraction of 25 neighbours
threshold-sampled labels
unrestricted labels
f
Acetylation K, + ESM-2, unrestricted test
Change in score percentile (points)
threshold training
unrestricted training
Merged annotation depth (bin median)